

Is a passing bay worth a transporter?

Fleet-network co-design: the road-vehicle exchange rate (T6), a retrofit menu that includes control granularity (H4), and vehicle type under road occupancy (H7)

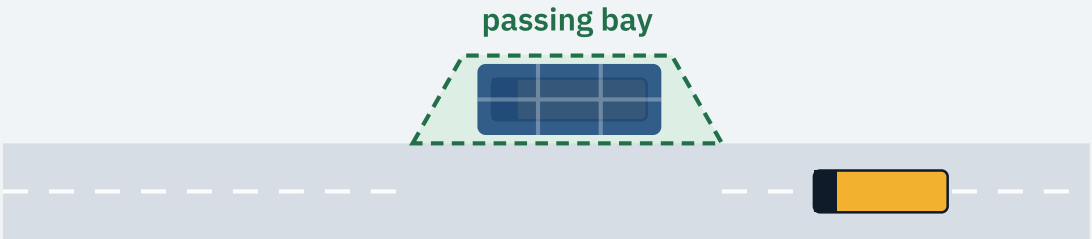
Is a passing bay worth a transporter? Fleet-network co-design for road-occupying transport (working title)

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Research manifesto v1: C4, H4, H7, T6 · a first probe in hand; the first exchange-rate table comes from T25

exchange rate = dual price ÷ daily capacity per vehicle



On a single lane, the loaded carrier pulls into the bay and an empty one passes

≈ ?



how many vehicles is it worth?

The sub-study in one minute

1

The question

To expand capacity where the network binds: buy vehicles or fix roads, large vehicles or small? We price each road change in vehicles.

2

The dividing rule

Plateau / T1' near 1: the bound is tight, its dual prices are the exchange rate, and fixing the road pays. Far below 1: the rule is the bottleneck, so change the orchestration first.

3

First evidence

Okpo main scenario (plateau 36% of T1'): changing road 161 moves reservation throughput by only +0.2% to +2.4%. Erection mix (87%): segmented blocking +10%, widening +4%, passing bays add nothing beyond segmentation.

4

Next

T23 delivers the route-free T1" and dual prices; T25 builds the first exchange-rate table (10 seeds, full K grid, in vehicles at whole-yard task volume); T26 tests H7 on floating-dock giant blocks.

Where the dock mouth dominates, a road that can be used in sections may be worth more than a wider one.

Chapter 5's question: when we spend to expand capacity, vehicles or roads, large vehicles or small?

When loaded vehicles hold the road exclusively, what determines the capacity of a shipyard block-transport system? How should fleet, orchestration and network be designed together?



What the chapter takes in

- **Chapter 3:** the bound $T1'$, the route-free $T1''$, their dual prices and binding elements
- **Chapter 4:** the best deadlock-free rule, which is the lower edge of the capacity interval and the inner evaluator
- **MY-B:** vehicle types, the team-transport model and vehicle prices from 33 tender awards

What the chapter delivers

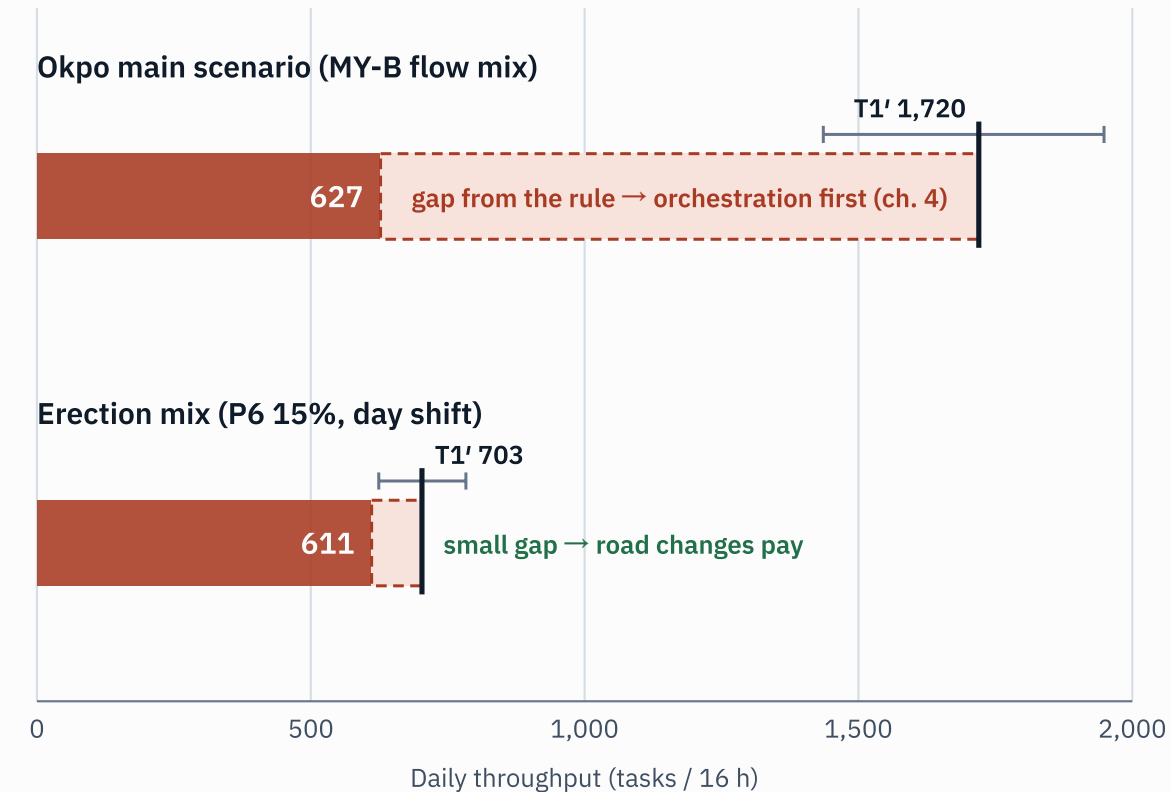
- The $C3/C4$ dividing rule and the first road–vehicle exchange-rate table
- A proof of $T6$ and critical-element identification; a retrofit menu with control granularity
- A bilevel co-design algorithm; the vehicle-type result ($H7$); upgrade values handed to chapter 6 for cross-yard prediction

The manifesto items this chapter serves: C4, H4, H7 and theorem T6

#	MANIFESTO TEXT (V1, 9 OCT 2026)	HOW THE CHAPTER DELIVERS IT
C4	Road–vehicle exchange rate: where the bound is tight, upgrading critical elements can replace vehicles, at a rate equal to the dual price converted to vehicles. When the plateau is far below the bound, change the orchestration first. This is the boundary between C3 and C4	The dividing rule is the chapter's backbone; the main scenario and the erection mix give one example each
H4	Hypothesis: where the network binds, one upgrade on a critical element is worth at least one vehicle; critical elements can be identified by dual prices. Upgrades to choose from: segmented blocking, dock-front holding bays, passing bays, one-way loops, widening, a second entrance. Criterion: the difference in vehicles needed before and after the upgrade in simulation, compared with the dual-price prediction. The two are expected to agree only when plateau / bound is close to 1. Holds: the exchange rate holds. Fails: the capacity bottleneck is orchestration, and chapter 4 is strengthened.	T25: 10 seeds, full K grid; difference in vehicles needed at whole-yard volume; dual predictions from T23
H7	Hypothesis: large vehicles occupy the road both loaded and empty; small ones can yield when empty, but teaming up for a lift costs extra empty runs. Case: multi-vehicle transport of giant blocks for floating docks. Holds: the optimal fleet mix changes with how strongly the network binds. Fails: vehicle type and network can be decided separately.	T26 floating-dock scenario; MY-B vehicle types and prices
T6	Road–vehicle exchange rate: given by the dual prices; only upgrades on binding constraints have value. For new links, use T1" Status: author	Proved by the author; T23 supplies T1" and the top-10 dual prices

C3 (the other side of the boundary): where the network binds, how much of the bound is reachable is decided by traffic orchestration: release order, waiting points, reservation mode, block granularity. Naive dispatching deadlocks.

The C3/C4 boundary: how far the plateau sits below the bound decides between orchestration and road changes



■ Whole-route reservation plateau (mean of $K = 100-150$) ■ Gap to T1'
 ■ T1' (whisker: range over 10 seeds)

Sources: thesis-core 83c16bf, results/t18_report_d10b9015cd4d/t18_summary.json (T18; Okpo, dock road 161 as one resource, kept as comparison from T22 on; saturated demand). Plateau = mean saturated throughput of whole-route reservation over $K = 100-150$ (10 seeds); the T1' whisker spans the 10 seeds. Erection mix = P6 at 15%, concentrated in the day shift (T18's stress scenario). Change readings from review appendix A (simulations 3 seeds). Rule: review §2.3; manifesto C1, C4.

36%

36% Far below 1: the rule is the bottleneck

Raising the bound does not move throughput; change orchestration first, as chapter 4 pushes the lower edge up. Example: on Okpo's main scenario three changes to road 161 lift T1' by 5–8% but reservation throughput by only 0.2–2.4%.

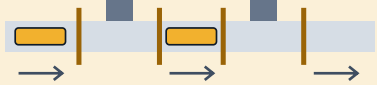
87%

87% Near 1: the bound is tight

Dual prices are the exchange rate; upgrading the binding element pays, so fix the road first. Example: in the erection mix, blocking road 161 in sections at its stops raises reservation throughput by 10%.

Capacity is an interval [best deadlock-free rule, T1']: chapter 4 pushes the lower edge up, chapter 3's tighter bounds press the upper edge down, and their ratio decides where chapter 5's money goes.

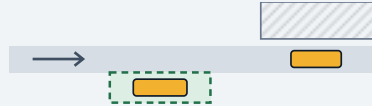
Retrofit menu: six changes, one purely in control granularity



CONTROL GRANULARITY

Segmented blocking (with direction locks)

A long road is cut at its stops into several resources; vehicles may follow in the same direction. Only the resource division changes, not the routes, so T1' still applies.



Dock-front holding bay

Vehicles waiting for the crane stand off the road and stop blocking the way to neighbouring stops. Where vehicles wait is not in the LP, so the value is read from simulation.



Passing bay

A wider stretch on a single lane where two vehicles pass. It changes routes, so the route-free T1'' prices it.



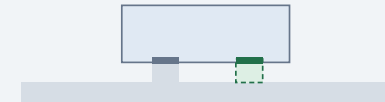
One-way loop

Turns head-on conflicts into same-direction following. Directions and shortest paths change, so T1'' is needed.



Widening

Raises the width capacity. Road 161 from 9 m to 12 m, say: two empties can pass, while a loaded carrier still mostly holds the road alone.



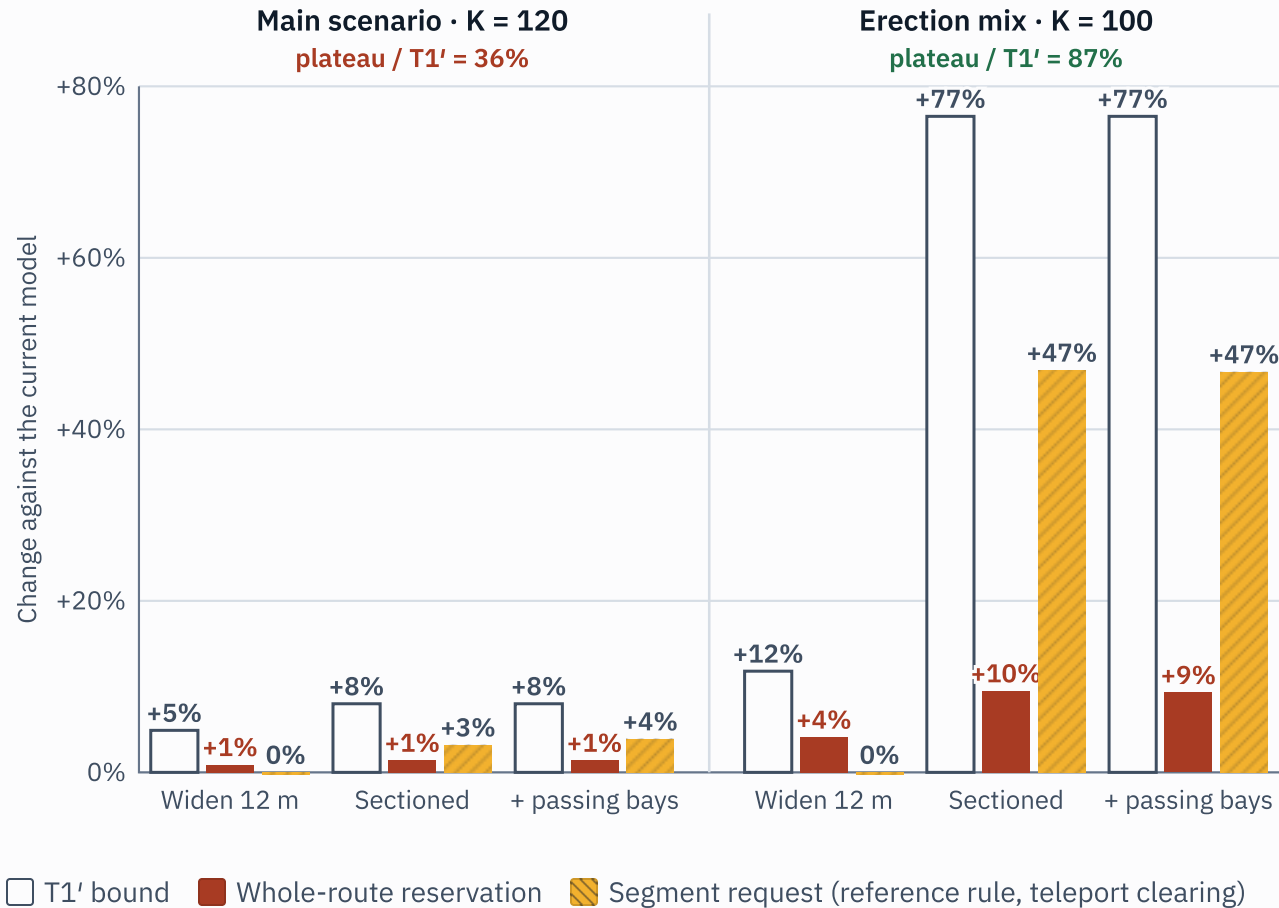
Second entrance

Adds a door where a door binds, sharing out the loading time held at the node. It adds a link, so T1'' is needed.

Control granularity is hidden capacity: segmented blocking costs no construction; it changes only how resources are divided and how vehicles are released. The manifesto lists block granularity under C3's orchestration and segmented blocking in H4's menu, so it sits right on the boundary.

Migration sources: railways' "shorter block sections raise line capacity" (blocking-time theory, UIC 406; exact references being checked in T24, to be verified); a mine's single-lane haulage with passing bays run as railway block sections (Eustace 2021, full text).

First evidence: in the main scenario road changes barely move throughput; in the erection mix, segmented blocking is worth more than widening



+0.2-2.4%

Main scenario: reservation throughput (K = 60, 120) after the three changes, while T1' moves by +5% to +8%. The rule limits throughput, not this road.

+10% / +4%

Erection mix, K = 100: reservation throughput after segmented blocking / widening; passing bays add nothing beyond segmentation.

+47% / 0

Segment-by-segment request in the same mix (reference rule, teleport clearing): segmentation / widening. No direction locks were set, so the gain includes teleport effects.

+3-15% / +3-5%

Seed by seed: segmentation / widening, same direction every time; T25 confirms the size with 10 seeds and the full K grid.

Sources: review (9 Oct 2026), appendix A and §§0.2, 2.3; the chart's numbers are parsed from the appendix A tables by make_data.py. Conditions: thesis-core 83c16bf code unchanged, road 161 altered on the built network; Okpo derived map rev 1825; saturated demand, 4 days; bounds 10 seeds, simulations 3 seeds (101-103), a prototype; the baseline is "whole road as one resource", and segmented blocking is Okpo's main scenario from T22 on (author's imagery check, 9 Oct 2026). Only the reservation changes can be used as numbers directly.

T6, the road-vehicle exchange rate: the bound's dual prices in vehicles; only binding elements are worth changing

$$\pi_r = \partial T1' / \partial H_r$$

extra tasks a day per extra hour a day on resource r

$$e_r = \pi_r \div (H / \bar{c})$$

in vehicles; $H / \bar{c}$ = one vehicle's daily capacity

- **Only upgrades on binding constraints have value:** a slack resource has a zero dual price.
- **With plateau / T1' near 1, e_r is the exchange rate;** far below 1, it is an upper bound on an upgrade's value.
- **Dual prices are marginal:** a large change moves the binding element (after sectioning in the main scenario it shifts from road 161 to road 010-6), so simulation checks it.
- **New links change routes:** passing bays, one-way loops and second entrances use the route-free T1'' (T23).
- **The H4 test:** dual prices rank and predict upgrade values; simulation then reads the difference in vehicles needed at whole-yard volume (T25). Where the two agree is the test of the C3/C4 rule.

The first reading (T18, Okpo, road 161 as one resource)

Scenario	Binding element (seeds)	Dual price tasks/day per h/day	In vehicles	Plateau / T1'
Erection mix	road 161 (9 m, 1,095 m) · 10/10	43.9	1.8	87%
	road 161 (9 m, 1,095 m) · 5/10	98.9	4.2	
Main scenario	road 010-6 (12 m, 168 m) · 4/10	115.5	4.9	36%
	road 026 (18 m, 348 m) · 1/10	118.3	5.0	

In the erection mix the bound is tight: one more hour a day on road 161 is worth about **1.8** vehicles, usable as the exchange rate. In the main scenario it is not tight: 4–5 vehicles is only an upper bound, which fits the probe, where changing road 161 moved reservation throughput by only 0.2–2.4%.

Sources: theorem as stated in the manifesto v1 theorem list, T6 (status: author); thesis-core 83c16bf, t18_summary.json (binding resources and dual prices, mean over the binding seeds out of 10) and the fleet bound in each batch's curves.csv ($H / \bar{c}$ = fleet bound ÷ K); docs/T18_report.md §2.2. The dual is degenerate: when several resources bind at once the price falls on one of them, hence the seed counts. Whole-yard volume of 300–600 a day: Yim 2008; Shen 2018 (review §2.1).

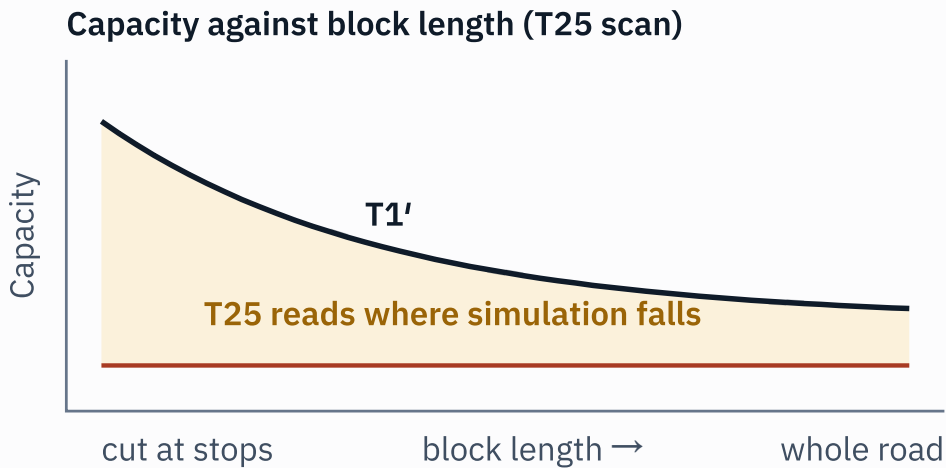
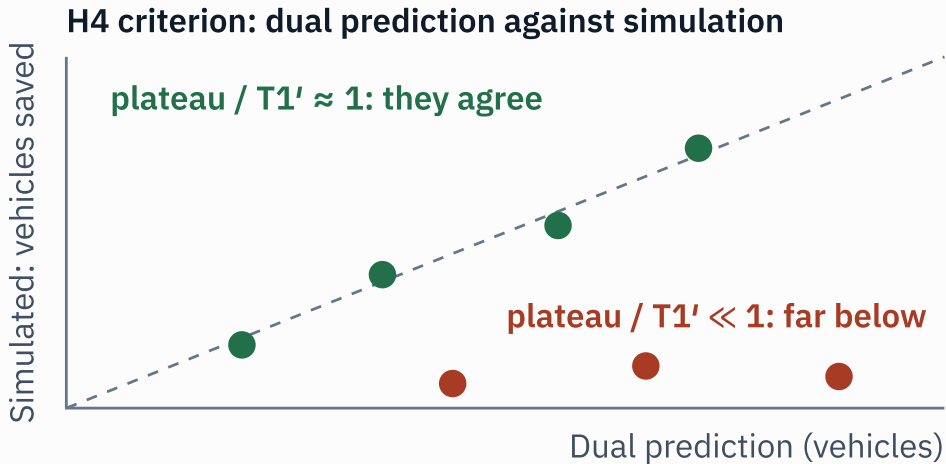
The first exchange-rate table (T25): each change in vehicles

CHANGE	PROTOTYPE: RESERVATION	DUAL PREDICTION (VEHICLES)	SIMULATION: Δ VEHICLES NEEDED
Segmented blocking (direction locks)	+10%	T23	T25
Dock-front holding bay	—	not seen by the LP	T25
Passing bays	0%	T23 · T1"	T25
Widening	+4%	T23	T25
Door granularity (Yantai building 014)	—	T25	T25

Prototype: appendix A, erection mix, K = 100, 3 seeds; the passing-bay row is “sections + bays” against sections alone. **T25:** 10 seeds, full K grid, whole-yard volume of 300 / 450 / 600 tasks a day; conclusions rest only on rules without teleport clearing.

Two more items in T25

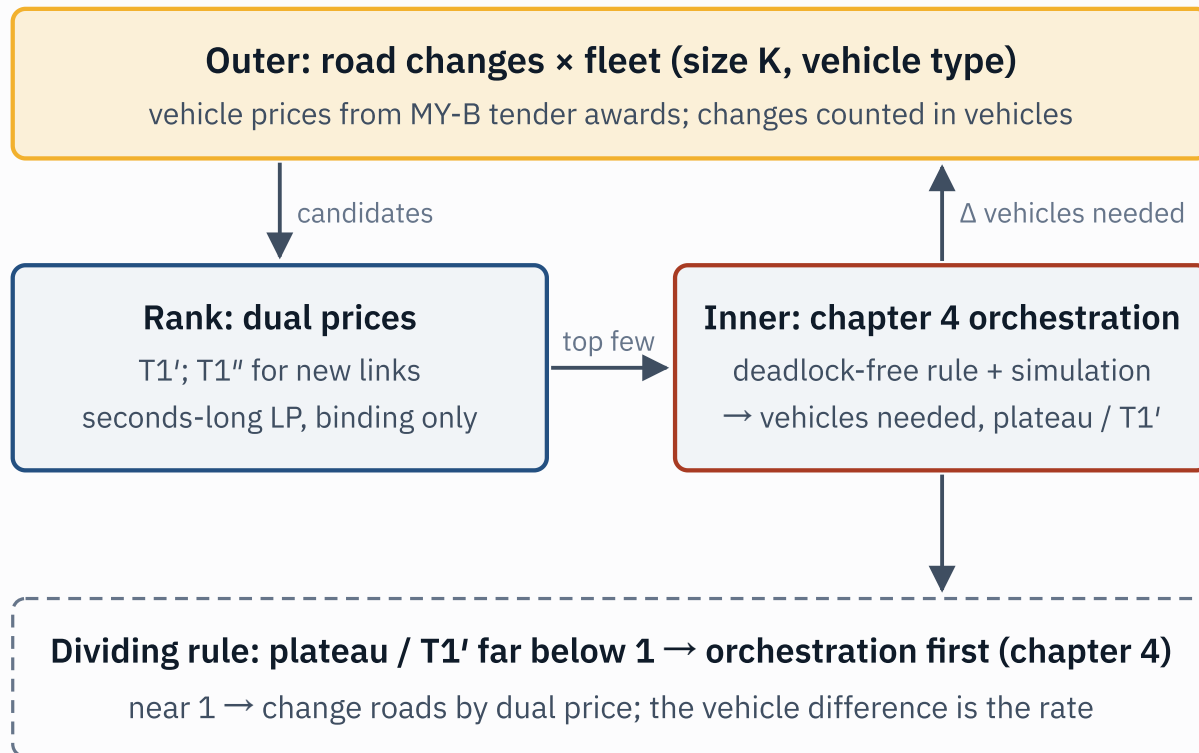
- **Dock three-resource network:** crane, single-lane dock road and one neighbouring stop; the threshold p^* against the dock road's spare passability, for the dock-mouth proposition (ch. 3).
- **Door granularity:** a door is held only while passing, loading moves to a separate bay, and doors with room to pass are shared by width; read Yantai's T1', binding element and plateau.



Both figures are schematics, not data

Sources: task board T25 (retrofit simulations and the first exchange-rate table; capacity vs block granularity, a block-length scan; door granularity; the dock three-resource network); review §2.3 and appendix A (prototype column read by make_data.py); manifesto H4 criterion and wording rule 2 (teleport rule as reference only). The two figures on the right are schematics, not data.

Bilevel co-design: changes and fleet outside, orchestration inside, dual prices to rank



Chapter 5's results ladder

FLOOR

Grid evaluation of fleets and road changes

TARGET

The T6 exchange rate and critical-link identification

CEILING

Bilevel co-design with optimality gaps

- Dual prices shrink the outer search to the binding elements, cutting the number of designs to simulate.
- Key-element review is a fixed step: the author checks the top three dual-price elements on the imagery, and the resource granularity is chosen from that before recomputing.

H7, vehicle type and occupancy: large vehicles occupy even when empty; small ones can yield but run extra empty trips to team up

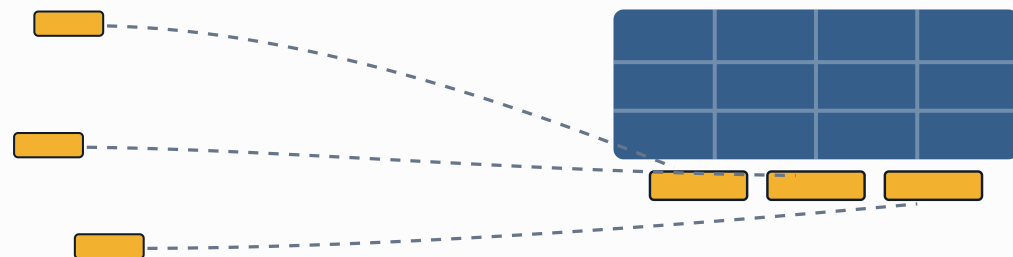
Large vehicle: holds the lane even when empty



Small vehicle: yields where passing is possible when empty



Small vehicles teamed: a giant block needs several, each running empty to meet



- **Where MY-B starts:** with static shortest paths and no traffic, one large enough vehicle is cheaper than any coupled team for every block; prices come from 33 public tender awards. H7 asks whether occupancy changes that answer.
- **The case (T26):** Okpo rev 1833 marks 3 floating docks, matching Hanwha Ocean's 4Q24 IR ("Floating Dock 3EA"). A floating dock is modelled as a quay-side sink plus a floating-crane resource with capacity 1–2 and a long cycle, reusing the crane code; floating-dock flow is a separate scenario.
- **Giant blocks:** assembled on land and carried by 2–4 coupled vehicles to the quay, wider and holding the road longer, with empty returns in convoy too. This is general practice knowledge; each item is sourced from public material before it enters the model (T24), and "about 10 giant blocks per ship" is to be verified.

[Open the fleet-sizing \(MY-B\) briefing →](#)

Established, and under way: what each task feeds

Established

- **A reliable bound:** T1' holds for whole-route reservation, segment request without teleporting, and any rule meeting (A1)–(A3); none of 17,760 occupancy simulation points on two yards exceeds it. The dual prices rest on it.
- **Binding elements can be located:** in the erection mix all ten seeds bind on road 161, one extra hour a day $\approx$ 1.8 vehicles (T18).
- **The dividing rule has two examples:** main scenario (36%) and erection mix (87%), consistent in direction (probe, 3 seeds).
- **Key-element review is a fixed step:** the checks of road 161 (neighbouring sites stay reachable $\rightarrow$ segmented is Okpo's main scenario) and Yantai's building 014 (two doors, vehicles can pass in front) both changed the model.

T22

running Physical rules (segment_safe deadlock avoidance, a no-clearing switch); dock road segmented with direction locks; whole-yard volume. $\rightarrow$ lower edge of the interval, base of the retrofit runs

T23

running System T1', the route-free T1'', the value of route choice, the top-10 dual prices. $\rightarrow$ the “dual prediction” column

T24

running External anchors and migration sources, including block granularity, floating docks and giant blocks. $\rightarrow$ literature for the menu and H7

T25

queued Retrofit simulations; capacity vs block length; door granularity; the dock three-resource network. $\rightarrow$ the first exchange-rate table, the prediction-vs-simulation figure, the granularity curve

T26

queued Okpo rev 1833 and the floating-dock scenario; finite holding area S and $K_0(S)$. $\rightarrow$ the H7 case section; the rule “what a holding area is worth in vehicles”

Risks and mitigations

RISK	MITIGATION
In the main scenario road changes barely move throughput	That is itself the C3-side reading: orchestration first; once chapter 4 raises the lower edge, read the exchange rate again at the new plateau / T1'
The probe has 3 seeds and no direction locks, and segment request is the reference rule (teleport clearing)	T25 uses 10 seeds and the full K grid; T22 adds direction locks and segment_safe; conclusions use only rules without teleporting
The dual is degenerate and only marginal	Report seed counts with prices; T23's system T1' removes sampling noise and gives the top 10; large changes are checked by simulation
New links mean the fixed-route T1' is no longer a bound	T23's route-free T1'' holds for any routing rule, and T1' – T1'' measures the value of route choice
Critical elements may be modelled conservatively (whole-road holding, exclusive loading at doors)	Key-element review as a fixed step; T25 adds door granularity; the whole-road model is reported alongside as comparison
Little public material on floating docks and giant blocks	T24 sources each item; floating-dock flow is a separate scenario, with synthetic parts labelled as scenarios

Candidate journals: several options, to be chosen once results are in

Is a passing bay worth a transporter? Fleet-network co-design for road-occupying transport

FIRST CHOICE

- *Transportation Research Part E*
- *IIE Transactions*

Exchange-rate table, dividing rule and bilevel algorithm: a logistics and industrial-engineering design problem

ALTERNATIVES (THEORY, OR)

- *Transportation Research Part B*
- *European J. Operational Research*
- *Omega*

Once the T6 and T1" duality results are strong, with weight on theorems and algorithms

APPLICATION-ORIENTED

- *Computers & Industrial Engineering*
- *Ocean Engineering*

Shipyard retrofit cases, floating-dock transport (H7) and the key-element review procedure

Chapter 3's first paper may also go to TR-E or IIE Transactions; the two papers will be spread across venues once T25's results are in.

References

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8. Yim, S.-B., Roh, M.-I., Cha, J.-H., Lee, K.-Y. (2008). Optimal block transportation scheduling considering the minimization of the travel distance without overload of a transporter. *J. Society of Naval Architects of Korea* 45(6), 646–655. doi:10.3744/SNAK.2008.45.6.646

Company material: Hanwha Ocean 4Q24 IR (Floating Dock 3EA), as recorded on the task board. Internal sources: manifesto v1 (thesis-core, tag manifesto-v1); the author's review (9 Oct 2026); the direction document, 2nd ed. (8 Oct 2026); T18 report and results at thesis-core 83c16bf.

Back to the whole thesis

Is a passing bay worth a transporter? First see how far the plateau sits below the bound.

<< 1

Plateau / T1' far below 1: the rule is the bottleneck; orchestration first.

≈ 1

Plateau / T1' near 1: the bound is tight, dual prices are the exchange rate, and fixing the road pays.

H7

Does occupancy change the choice between large vehicles and teams?
Tested on floating-dock transport.

← Back to the thesis briefing

Occupancy model and bounds →

Capacity interval →

Traffic orchestration →

Multi-yard test →

Fleet sizing (MY-B) →

Thank you.